ASKIY ENERGY
"COMMUNITY DRIVEN"

Initial Draft Proposal
P.A.P.E.R.
Planning Automation for Projects: Energy and Retrofits

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1. Overview

Project Name:

PAPER: **P**lanning **A**utomation for **P**rojects, **E**nergy, and **R**etrofits

Primary Applicant:

Askiy Energy

PAPER: Empowering Indigenous Energy Futures in Fort Chipewyan and Beyond

The PAPER platform is an initiative rooted in Indigenous values and perspectives, addressing the systemic challenges faced by First Nations communities in the energy retrofit sector. By prioritizing Indigenous data sovereignty and community-led solutions, PAPER represents a significant step forward in creating equitable access to energy efficiency tools for communities with limited capacity.

PAPER will be a solution and a Holistic Approach from an Indigenous perspective, energy planning is not just about technology—it's about preserving the interconnected relationship between people, land, and resources. PAPER reflects this holistic worldview by:

1. Respecting Indigenous Data Sovereignty: Data collection and utilization in PAPER are guided by OCAP® principles (Ownership, Control, Access, and Possession), ensuring that communities retain control over their information and its use aligns with cultural

protocols. This enables governance bodies and members to have a peace of mine and freedom to trust PAPER with their intellectual property.

2. Community Empowerment: Through its AI Education Program, PAPER equips local teams with the knowledge and skills to independently plan and implement energy retrofits. This approach aligns with traditional Indigenous values of capacity building and self-determination, fostering long-term resilience. While adapting and being resilient to climate change.

3. Scalable, Modular Design: The tool is adaptable, recognizing that no two communities or municipalities are alike. PAPER's design considers the diverse realities of First Nations across Alberta, Saskatchewan, and Manitoba, and beyond ensuring scalability without sacrificing cultural or regional relevance.

Local Impacts: Fort Chipewyan, with its legacy as one of the oldest settlements in Alberta, has experienced firsthand the challenges of energy dependency and climate change. With its remote location, high costs of diesel dependence, and limited local capacity for large-scale energy projects, Fort Chipewyan embodies the challenges Askiy Energy and PAPER seeks to address.

1. Building on Proven Success

Fort Chipewyan's successful deployment of a 2.34 MW solar and 1.5 MW battery system—the largest off-grid project of its kind in Canada—demonstrates the transformative power of Indigenous-led clean energy initiatives. PAPER leverages this success, using automation and AI to replicate similar achievements in other communities.

2. Addressing Capacity Gaps

Many First Nations, like Fort Chipewyan, face staffing shortages and technical knowledge barriers that delay energy projects. PAPER simplifies the process by automating data collection, analysis, and funding documentation, removing critical bottlenecks for communities with limited capacity.

3. Cultural and Environmental Stewardship

For Indigenous communities, energy retrofits go beyond efficiency—they are about restoring balance. PAPER ensures that energy projects align with traditional teachings by emphasizing sustainability and minimizing environmental harm.

Broader Regional Benefits

For First Nations across the Prairie Provinces, PAPER provides a pathway to:

Reduce Dependency on Diesel: Many remote Indigenous communities rely heavily on diesel power, which is both expensive and environmentally damaging. PAPER streamlines the transition to clean energy systems, helping reduce emissions and lower energy costs.

Access Funding Opportunities: By automating grant applications and funding documentation, PAPER bridges the gap between Indigenous communities and much-needed financial resources.

Strengthen Collaborative Networks: The platform fosters connections between communities, sharing success stories like Fort Chipewyan's solar project to inspire others.

Vision for the Future

PAPER is more than a tool—it's a catalyst for self-determination, It empowers Indigenous communities to lead their own clean energy transitions, creating solutions that are culturally relevant, environmentally sustainable, and economically viable. With Fort Chipewyan as an anchor example, PAPER has the potential to redefine how energy projects are envisioned, planned, and executed across Turtle Island.

Purpose:

PAPER is an innovative platform designed to revolutionize energy retrofit planning for Indigenous communities across Alberta, Saskatchewan, and Manitoba. By automating data collection, analysis, and document creation, PAPER addresses the unique challenges of capacity limitations, data underutilization, and cost barriers. It empowers communities to streamline retrofit planning, access funding opportunities, and foster long-term energy efficiency.

Through modular design and AI integration, PAPER enables scalable and customizable energy planning solutions while respecting Indigenous data sovereignty. The platform also includes an **AI Education Program** to train community teams in leveraging the tool effectively and independently.

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2. Details of the Grant

Funding Program:

Regional Artificial Intelligence Initiative (RAII) - Prairies Economic Development Canada

- **Funding Amount:** $3,000,000 - $5,000,000 over 3 years
- **Program Objective:** To support projects that use AI technologies to enhance economic opportunities, innovation, and community development in Alberta, Saskatchewan, and Manitoba.

Potential Organizational Structures for PAPER

PAPER funding approach can be structured as either a **For-Profit Indigenous-Owned Company** or a **Non-Profit Entity**. Each structure has distinct advantages and disadvantages, particularly in securing funding, licensing, and ownership considerations.

Option 1: For-Profit Indigenous-Owned Company

Under this model, PAPER would operate as an Indigenous-owned for-profit business, ensuring **full control and ownership of the software** by Askiy Energy.

Funding Considerations

- **PrairiesCan Contribution:** If at least **51% Indigenous-owned**, up to **80% of project costs** can be funded as a **repayable contribution**.
- If the project benefits **the broader Indigenous community**, rather than just the company, **50% of costs** could be covered as a **non-repayable contribution**.
- **Stacking:** Government assistance is **capped at 50%** for eligible capital costs and **75%** for other eligible costs.
- **In-Kind Contributions:** The remaining **50% contribution** can include in-kind contributions (e.g., from the University of Waterloo or other partners).

Pros

✅ **Greater Flexibility & Control:**

- Askiy retains **100% ownership** of the software and all intellectual property.
- The ability to license the software to other communities or organizations.
- **Revenue generation opportunities** (e.g., SaaS model or consulting services) without limitations.

✅ **Stacking Funding Possibilities:**

- Ability to combine multiple funding sources, including **Indigenous Services Canada (ISC), climate and energy funding, private investment, and in-kind contributions**.

✅ **Business Development Funding Alignment:**

- Since this is a **Business Development grant**, the likelihood of securing funding as a **for-profit** entity may be higher.

Cons

❌ **50% Contribution Must Be Secured Before Applying:**

- All non-PrairiesCan financing **must be confirmed upfront**, including any **private investment or in-kind contributions**.
- Future revenue from licensing or operations **cannot** be counted as matching funds.
- This may pose a **significant challenge** in securing capital **before** the application process.

❌ **Repayable Contribution Requirement:**

- Ideally we will be accepted under "Project that Benefits the community" so 30% is repayable, 50% is not.
- If deemed that no clear benefit to community than the **80% PrairiesCan contribution** is **repayable over 5 years**, starting **one year after project completion**.
- **Cash flow planning is critical** to ensure loan repayment without financial strain.

Option 2: Non-Profit Structure

In this model, PAPER would be developed under an **existing or new non-profit entity**, which would **own the software** and have the ability to **license it** to other groups, including Askiy Energy.

Funding Considerations

- **PrairiesCan Contribution:** Up to **80% of project costs** as a **non-repayable grant**.
- **Stacking:** If the non-profit lacks sufficient resources, **government assistance can cover up to 100% of eligible costs**.
- **Ownership & Licensing:** The **non-profit must retain ownership** of all assets funded by PrairiesCan.
- **Partnership Approach:** A potential structure could involve **Three Nations Energy** or another Indigenous-led non-profit **partnering with Askiy Energy**.
- **Askiy Energy** starts a **new Non-Profit entity** (this could take time)

Pros

✅ **80% of Costs Are Non-Repayable:**

- No requirement to **repay funds**, reducing long-term financial burden.
- Less pressure to generate immediate revenue post-project.

✅ **Potential for Full Funding (100% Stacking):**

- If additional funding sources (federal, provincial, or municipal) are secured, **the project could be fully funded without requiring private capital or loans**.

✅ **Lower Upfront Capital Requirements:**

- Unlike the for-profit model, **funding does not need to be secured in advance**, making it **easier to apply and qualify**.

Cons

❌ **Complex Ownership & Licensing Considerations:**

- The **non-profit retains ownership** of the software, **not Askiy Energy**.
- The software **could be licensed back** to Askiy Energy, but this adds complexity.

❌ **Longer Approval Process & Bureaucracy:**

- Working within a non-profit framework requires **additional approvals and oversight**, which **may slow down implementation**.
- Decision-making could involve a **board of directors**, adding administrative overhead.

❌ **Lower Likelihood of Success with Business Development Funding:**

- Since **PrairiesCan primarily funds business development projects**, a **for-profit** model may have a **higher chance of being approved**.

Recommended Next Steps

Given the strengths and challenges of both models, the decision should be based on:

1. **Securing the required 50% match funding** – If Askiy and partners can confirm in-kind and cash contributions upfront, the **For-Profit model** may be preferable.
2. **Determining ownership priorities** – If maintaining **100% control over licensing and intellectual property** is essential, the **For-Profit model** is ideal.
3. **Balancing funding availability and application success** – If obtaining the **full 80% as a non-repayable contribution** is a priority and complexity is manageable, the **Non-Profit model** should be considered.

By carefully weighing these factors, Askiy Energy and its partners can **select the best path forward** to maximize funding, retain ownership, and ensure long-term sustainability for PAPER.

Hypothetical Scenario: For-Profit Stream Benefiting Indigenous Communities

Total Project Cost: $3,000,000

- **PrairiesCan Contribution (80% of project costs = $2,400,000)**
  - **Non-Repayable Portion (50%): $1,500,000**
  - **Repayable Portion (30%): $900,000** (0% interest, repaid over 5 years, starting 1 year after project completion)
- **Remaining 20% of Project Costs ($600,000) Covered By:**
  - **Indigenous Services Canada (ISC) - Housing & Energy Funding: $250,000**
  - **Indigenous Off-Diesel Initiative (IODI) - Renewable Energy Transition: $200,000**
  - **3rd Party Contributions (Private/Philanthropic Sources): $100,000**

- **In-Kind Contributions from Project Partners (University of Waterloo, Industry Experts, Technical Support, Data Analysis): TBD** (Could offset a portion of the 20% contribution)

Project Overview: PAPER – Indigenous-Led Energy Retrofit & AI Planning Platform

PAPER will be an **AI-powered energy planning platform** designed to **empower Indigenous communities** in Alberta, Saskatchewan, and Manitoba by automating energy retrofit planning, funding access, and project execution. (After the pilot phase the tool can be implemented nationally)

The project will enable communities to:
✅ **Transition away from diesel reliance** through streamlined clean energy planning.
✅ **Secure funding for energy retrofits** using AI-driven applications and automated compliance tracking.
✅ **Support local Indigenous businesses** by generating contractor-ready retrofit plans.
✅ **Foster Indigenous data sovereignty** by ensuring communities retain control over their energy data.

Breakdown of Funding Allocation

Category	Budget Allocation ($)	Funding Source
Software Development & AI Integration	$2,200,000	**PrairiesCan (50% non-repayable, 30% repayable)**
Community Pilot Programs (5-10 Indigenous Communities)	$200,000	**PrairiesCan, ISC, IODI**
Capacity Building & Training (AI Education Program for Indigenous Technicians)	$200,000	**ISC, In-Kind Contributions**
Deployment of Retrofit Planning in 10+ Communities	$200,000	**PrairiesCan, 3rd Party, In-Kind**
Licensing, Legal & Compliance (Ensuring 100% Indigenous Ownership via Askiy Energy)	$200,000	**PrairiesCan (Repayable), Private Capital**
Total	**$3,000,000**	**80% PrairiesCan + 20% Stackable Funding**

Key Benefits to Indigenous Communities

✅ **Scalable Impact:** A **centralized AI-driven energy planning platform** that can **be used across multiple Indigenous communities** to streamline energy transitions and retrofits.

✅ **Indigenous Economic Development:** The **software is 100% Indigenous-owned by Askiy Energy**, ensuring that all future revenue stays within Indigenous businesses and communities.

✅ **Increased Energy Independence:** Communities will be able to **leverage AI-generated reports** to apply for additional funding, secure contractor quotes, and plan retrofit projects without external reliance.

✅ **Employment & Capacity Building:**

- AI-driven energy audits and training programs will create **job opportunities for Indigenous technicians, consultants, and energy auditors**.
- Communities will be trained to use and adapt the platform to their **specific regional needs**.

✅ **Government & Private Investment Alignment:**

- By leveraging **PrairiesCan, ISC, and IODI funding**, PAPER ensures **long-term sustainability and alignment with Canada's Indigenous Clean Energy initiatives**.
- The ability to **stack additional funding** ensures communities are not financially burdened while benefiting from long-term cost savings through **clean energy transitions**.

Risk Mitigation & Financial Sustainability

- **Ownership & Licensing:** Askiy Energy retains **100% ownership of the platform**, ensuring control over future **commercial licensing opportunities**.
- **Revenue Model:**
  - **Licensing to additional Indigenous communities** for a **subsidized fee** ensures long-term financial sustainability.
  - **Partnerships with Indigenous businesses** (e.g., solar providers, energy auditors) will help **generate revenue** without requiring additional external funding.

- **Repayment Terms Management:** Since **only 30% ($900,000) is repayable**, structured revenue streams from licensing and energy savings reinvestment **will cover repayment obligations without burdening operations**.

Why This Model Works Best

✅ **Maximizes Grant Opportunities** – The **50% non-repayable contribution** minimizes financial risk.
✅ **Ensures Indigenous Ownership & Control** – No loss of ownership to third parties.
✅ **Supports Economic Self-Sufficiency** – Enables revenue generation through licensing and partnerships.
✅ **Aligns with Business Development Mandate** – Higher likelihood of success due to clear economic benefits.

Conclusion

This **For-Profit Indigenous-Owned Model** allows PAPER to operate **sustainably**, scale its **impact across multiple Indigenous communities**, and create long-term **economic, environmental, and social benefits**. By leveraging **stackable funding sources**, Indigenous communities will gain access to a **cutting-edge energy planning tool** while ensuring that ownership and economic opportunities remain **within Indigenous hands**.

3. The Challenge Statement

Key Issues:

1. **Limited Capacity in Communities**:
    - Many Indigenous communities lack the internal workforce, expertise, and resources to manage data collection, planning, and documentation independently.
    - Competing priorities among staff often delay critical energy initiatives.
2. **Underutilized Data**:
    - Existing energy data, utility records, and housing information are often not analyzed due to a lack of tools and expertise.

    - Without aggregated and validated data, creating actionable energy retrofit plans remains a significant hurdle.

3. **High Costs and Scalability Barriers**:
    - Piecemeal retrofitting approaches are costly and inefficient.
    - Bulk procurement and centralized planning require advanced tools and detailed documentation that many communities cannot produce.
4. **Knowledge Gaps in AI Technology**:
    - Communities often lack familiarity with AI-powered tools, limiting their ability to leverage advanced automation for energy planning.

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4. Project Objectives

Primary Goals:

1. **Automate Planning Processes**:
    - Streamline data collection, analysis, and document generation to create high-quality retrofit plans, RFQs, and financial models.
2. **Empower Indigenous Communities**:
    - Train local teams through the AI Education Program, fostering autonomy in managing retrofit projects.
3. **Promote Scalability and Cost Efficiency**:
    - Support aggregated planning and procurement, reducing costs and improving implementation timelines.
4. **Ensure Data Sovereignty**:
    - Develop and implement robust policies ensuring communities retain ownership and control over their data.
5. **Validate and Scale**:
    - Demonstrate the effectiveness of PAPER through pilots in 5-10 communities, creating a replicable model for future use.

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5. Pilot Community Journey

The pilot will involve **5 Indigenous communities** in Alberta, Saskatchewan, and Manitoba.

1. **Fort Chipewyan** *(Alberta)*
2. **Lac La Ronge Indian Band** *(Saskatchewan)*
3. **Frog Lake First Nation** *(Alberta)*
4. **Pinaymootang First Nation** *(Manitoba)*
5. **Ahtahkahkoop Cree Nation** *(Saskatchewan)*

Step-by-Step Journey:

1. **Community Engagement**:
    - Conduct initial meetings to introduce PAPER and its benefits.
    - Define project scope, identify community priorities, and formalize a **Data Access Agreement**.
    - Deliverable: Signed agreements and recorded meeting minutes.
2. **Data Collection**:
    - Perform energy audits for varied housing archetypes.
    - Collect key data, including utility usage, building characteristics, and occupancy patterns.
    - Deliverable: Comprehensive dataset uploaded to the PAPER platform.
3. **Data Input and Validation**:
    - Input data into the PAPER platform.
    - Validate and analyze data using integrated tools like Volta Snap and CP Logic's Utility Bill Analysis Tool.
    - Deliverable: Initial dashboard summarizing energy usage and retrofit opportunities.
4. **Plan Generation**:
    - Use PAPER to generate **Actionable Retrofit Plans (ARPs)** and **Scopes of Work (SOW)** templates.

    - Deliverable: Draft ARPs and RFQs customized to each community.

5. **Community Feedback and Adaptation**:
    - Present findings to the community and collect feedback through AI-guided surveys.
    - Adapt plans based on community input, ensuring alignment with local values and priorities.
    - Deliverable: Finalized ARPs reflecting community feedback.
6. **Procurement and Financial Planning**:
    - Upload contractor quotes, compare them against the SOW, and finalize a detailed financial plan.
    - Deliverable: Comprehensive financial documentation, including projected cost savings.
7. **Implementation and Monitoring**:
    - Track project progress using automated reporting tools.
    - Deliverable: Final implementation report and ongoing measurement and verification plan.

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6. AI Education Program

Goal: Enable community teams to independently use and maintain PAPER, ensuring sustainable energy planning.

Program Structure:

1. **Initial Training**:
    - Onboarding workshops covering data input, validation, and document generation.
2. **Ongoing Support**:
    - AI-guided tutorials, helpdesk support, and regular webinars.
3. **Advanced Modules**:
    - Training in customizing templates and analyzing trends.

Deliverables:

- Certified training programs for community members.
- User manuals tailored to local needs.

7. Roles and Responsibilities

A. Askiy Energy

Askiy Energy Corp, a 100% Indigenous-owned and operated renewable energy company based in Doghead, Mikisew Cree Reserve near Fort Chipewyan, Alberta, provides sustainable energy solutions that benefit both clients and the environment. Their services include solar power installation, community energy planning, energy efficiency audits, and smart grid implementation.

Scope and Contribution to PAPER

In the PAPER project, Askiy Energy will leverage its expertise to lead community engagement, capacity building, and the implementation of renewable energy solutions. Their specific contributions include:

1. **Community Engagement and Coordination**:
    - **Initial Outreach**: Engage with participating Indigenous communities to introduce the PAPER project, outline its benefits, and establish collaborative relationships.
    - **Data Access Agreements**: Facilitate the development and signing of agreements that define project scope, data collection methodologies, and data usage policies, ensuring alignment with community priorities.
    - **Ongoing Communication**: Maintain regular contact with community stakeholders to provide updates, gather feedback, and address concerns throughout the project lifecycle.
2. **Capacity Building and Training**:
    - **AI Education Program Delivery**: Lead the implementation of training workshops and provide ongoing support to community teams, enabling them

to effectively utilize the PAPER platform and manage energy retrofit projects independently.

- **Resource Development**: Create tailored training materials, user manuals, and support resources that cater to the specific needs and contexts of each community.

3. **Renewable Energy Assessments and Planning**:
   - **Energy Efficiency Audits**: Conduct comprehensive audits to identify areas for improvement, aiming to reduce energy consumption and increase savings for clients.
   - **Community Energy Planning**: Develop detailed plans that identify opportunities for energy efficiency and clean energy solutions, considering infrastructure, land use, and future growth to achieve community energy goals.
4. **Implementation of Renewable Energy Solutions**:
   - **Solar Power Installation**: Leverage experience from leading Canada's largest solar farm in a remote community to design and implement solar energy systems tailored to each community's needs.
   - **Smart Grid Implementation**: Design and implement smart grid systems that optimize energy distribution, enhancing reliability, efficiency, and integration of renewable energy sources.
5. **Cultural Integration and Community Programs**:
   - **Youth and Elder Engagement**: Develop programs that connect generations, fostering the transmission of knowledge, culture, language, and traditions related to energy practices.
   - **Local Food Production Initiatives**: Support projects like the Grocer SEACAN to promote local food production, creating employment opportunities and reducing reliance on external sources.

Askiy Energy's deep-rooted commitment to sustainable energy solutions, combined with its cultural understanding and community engagement expertise, positions it as a pivotal partner in the PAPER project. Their contributions will ensure that the project is implemented in a manner that is both technically effective and culturally resonant, empowering Indigenous communities to achieve their energy sustainability goals.

B. CP Logic

CP Logic, founded by Cedric Pepelea, is a consultancy dedicated to advancing sustainable energy solutions across Canada. The firm specializes in scaling energy efficiency programs, developing community energy financing mechanisms, and providing transformative training methods that engage communities in STEM and sustainability initiatives.

Scope and Contribution to PAPER

In the PAPER project, CP Logic will leverage its expertise to oversee project management, strategic planning, and community engagement. Their specific contributions include:

1. **Project Management and Strategic Planning**:
    - **Timeline Oversight**: Develop and monitor detailed project timelines to ensure all phases of the PAPER initiative progress as scheduled.
    - **Risk Assessment**: Identify potential challenges and implement mitigation strategies to maintain project momentum and integrity.
    - **Reporting**: Prepare comprehensive reports detailing project milestones, outcomes, and areas for improvement, ensuring transparency and accountability.
2. **Community Engagement and Capacity Building**:
    - **Energy Coaching**: Utilize experience from conducting over 300 one-on-one energy coaching sessions to educate community members on energy efficiency practices and the benefits of retrofit programs.
    - **Training Programs**: Develop and deliver training workshops aimed at building local capacity to manage and sustain energy projects, fostering community autonomy.
3. **Development of Analytical Tools**:
    - **Utility Bill Analysis Tool**: Create and implement a tool designed to analyze utility bills, providing insights into energy consumption patterns and identifying opportunities for efficiency improvements.

- **Data Validation and Verification**: Ensure the accuracy and reliability of collected data, forming a solid foundation for informed decision-making within the PAPER platform.

4. **Strategic Partnerships and Scaling**:
    - **Collaboration**: Leverage existing relationships with organizations to facilitate the scaling of energy solutions across Canada, ensuring the PAPER project's methodologies can be replicated and adapted in various contexts.
    - **Resource Mobilization**: Assist in securing additional resources and support to enhance the project's reach and impact.

CP Logic's extensive experience in sustainable energy initiatives, project management, and community engagement positions it as a crucial partner in the PAPER project. Their contributions will ensure the project is executed efficiently, effectively, and in a manner that empowers Indigenous communities to achieve their energy sustainability goals.

C. Parachute Software

Parachute Software, established in 2005, specializes in delivering configurable software solutions for energy efficiency programs. Their flagship platform, DropZone, has been instrumental in managing some of Canada's most ambitious energy efficiency initiatives, including GreenON and Energy Efficiency Alberta.

Scope and Contribution to PAPER

In the PAPER project, Parachute Software will leverage its extensive experience to develop and integrate a comprehensive platform tailored to the needs of Indigenous communities. Their contributions include:

1. Platform Development and Customization:

- Design and Build: Develop a user-friendly, modular platform that automates data collection, analysis, and document generation for energy retrofit planning.

- Configurability: Ensure the platform is adaptable to various community requirements without necessitating extensive IT upgrades.

2. System Integration:

- Third-Party Tools: Integrate essential tools such as Volta Snap and CP Logic’s Utility Bill Analysis Tool to enhance data validation and analysis capabilities.

- External Systems: Utilize DropZone's flexible architecture with APIs to connect to external systems, facilitating seamless data exchange and preventing data silos.

3. Data Management and Reporting:

- Real-Time Analytics: Implement features that allow for real-time data analysis and reporting, utilizing Parachute's expertise in actionable analytics and Microsoft Power BI integrations.

- Comprehensive Tracking: Develop functionalities to track every data point and user action, ensuring detailed program reporting and effectiveness assessment.

4. Mobile Application Development:

- DropZone Mobile: Create a fully integrated mobile application to facilitate in-field data collection, measure installation, photo uploads, and customer signature capture, enhancing the platform's accessibility and usability.

5. User Training and Support:

- AI Education Program: Collaborate in delivering training workshops and ongoing support to community teams, ensuring they can effectively utilize the platform and maintain autonomy in managing energy retrofit projects.

Parachute Software's proven track record in deploying configurable, integrated solutions for large-scale energy efficiency programs positions them as a pivotal partner in the PAPER project. Their expertise will ensure the platform is robust, adaptable, and tailored to meet the specific needs of Indigenous communities, thereby facilitating successful energy retrofit planning and implementation.

D. Volta Research

Volta Research, a not-for-profit organization, advances knowledge-based decision-making for low-carbon technologies by supporting research processes and providing analytical expertise. Their services encompass emissions accounting, solar energy solutions, green building initiatives, and smart grid development.

Scope and Contribution to PAPER

In the PAPER project, Volta Research will leverage its expertise to enhance data integration and analysis, focusing on energy assessments and retrofit recommendations. Their specific contributions include:

1. **Energy Assessment and Modeling**:
    - **Development of Analytical Tools**: Create user-friendly software applications that streamline energy assessments and modeling for existing low-rise residential buildings, integrating on-site data collection with industry-standard energy modeling software like HOT2000.
    - **Enhanced Analytics**: Provide location-specific energy and emissions analytics to offer tailored retrofit recommendations, aiding Certified Energy Advisors in delivering effective solutions.
2. **Data Integration**:
    - **Volta SNAP Platform**: Utilize and customize the Volta SNAP platform to facilitate seamless data collection and analysis, ensuring compatibility with existing energy modeling tools and standards.
    - **Integration with PAPER**: Ensure that data from energy audits and assessments are effectively integrated into the PAPER platform, enabling comprehensive analysis and reporting.
3. **Training and Capacity Building**:
    - **Educational Support**: Assist in developing training materials and workshops to educate community teams on using energy assessment tools and interpreting data for informed decision-making.
    - **Ongoing Support**: Provide continuous assistance to community teams, ensuring they can effectively utilize the tools and methodologies introduced.
4. **Research and Development**:

- **Innovative Solutions**: Collaborate on developing new tools and methodologies that address specific challenges identified during the project, such as affordability and accessibility of energy retrofits.
- **Pilot Testing**: Participate in pilot projects to test and refine tools and approaches, ensuring they meet the needs of Indigenous communities involved in the PAPER project.

Estimated Cost Contribution: $400,000 (13.3% of the total project budget)

Volta Research's extensive experience in energy assessments, modeling, and low-carbon technology solutions positions them as a vital partner in the PAPER project. Their contributions will ensure that the platform provides accurate, actionable insights, empowering Indigenous communities to make informed decisions about energy retrofits and sustainability initiatives.

E. University of Waterloo – Future Cities

University of Waterloo Future Cities Institute (FCI): Scope and Contribution to PAPER

The **Future Cities Institute (FCI)** at the University of Waterloo is a premier interdisciplinary research hub committed to developing innovative solutions for sustainable urban development. FCI focuses on creating tools and methodologies for healthy, prosperous, and environmentally sustainable communities. A core area of their expertise includes the development of AI-driven tools for community energy planning and policy design, aligning perfectly with the objectives of the PAPER project. (uwaterloo.ca)

Scope and Contribution

The University of Waterloo's FCI will play a critical role in ensuring the success of PAPER through the following contributions:

1. **Support AI Development with Software Partners**

   Collaborate with leading software partners to support the development and integration of advanced AI tools tailored for the PAPER platform. These partnerships will enhance the platform's capabilities, ensuring cutting-edge technology is deployed to address the unique energy planning needs of Indigenous communities.

2. **Manage Quality Assurance of PAPER Output**

   Implement rigorous quality assurance (QA) processes to ensure the accuracy and reliability of PAPER's outputs. This includes providing reinforcement learning and continuous feedback mechanisms to fine-tune the AI models, ensuring they remain aligned with best practices and evolving community requirements.

3. **Program Validation**: Conduct rigorous studies to evaluate the methodologies and outputs of the PAPER platform, ensuring alignment with best practices in energy planning and policy.

4. **Case Study Development**: Collaborate with pilot communities to document the implementation of PAPER, highlighting challenges, successes, and opportunities for improvement.

5. **Scalability Analysis**: Assess the feasibility of scaling the platform to other communities and regions, providing recommendations for adaptation and expansion.

- **Interdisciplinary Collaboration**:

  1. **Cross-Sector Expertise**: Leverage FCI's expertise in urban planning, energy systems, AI development, and sustainability to enhance PAPER's capabilities.

  2. **Partnership Facilitation**: Foster connections between academia, Indigenous communities, government agencies, and industry partners to create a cohesive, collaborative ecosystem for the project.

- **Capacity Building and Talent Development**:

  1. **Training Programs**: Design and deliver training programs for Indigenous communities to empower local teams with the skills to use PAPER and the integrated AI tools effectively.

  2. **Student Engagement**: Involve university students and researchers in the project, providing hands-on experience in real-world energy planning and sustainability initiatives.

Leadership Expertise

The project will benefit greatly from the leadership of **Eryn Stewart**, Managing Director of FCI. Eryn was the **Founding Managing Director of Indigenous Clean Energy (ICE)**, where she developed and implemented innovative programming to advance clean energy projects in Indigenous communities across Canada. Her experience includes:

- Deep knowledge of Indigenous programming and funding mechanisms.
- Proven success in facilitating meaningful engagement with Indigenous communities.
- Expertise in ensuring clean energy initiatives align with cultural, social, and economic goals.

Eryn's leadership will ensure that the PAPER project is culturally sensitive, community-driven, and aligned with Indigenous aspirations for sustainable development.

Estimated Cost Contribution

Activity	**Cost ($)**
Support AI Development with Software Partners	$75,000
Manage Quality Assurance of PAPER Output	$250,000
Program Validation	$75,000
Total	**$400,000**

Conclusion

The University of Waterloo's Future Cities Institute will be a pivotal partner in the PAPER project. Their cutting-edge AI tools for Community Energy Planning, combined with their expertise in interdisciplinary research, capacity building, and cultural sensitivity, will significantly enhance the project's success. This partnership ensures that PAPER is not only technically robust but also strategically aligned with the unique needs and goals of Indigenous communities, fostering long-term sustainability and resilience.

8. Cost Estimated Breakdown Table

Partner	Role	Responsibilities	Estimated Cost ($)
Askiy Energy	Lead Applicant, Community Engagement	Train local teams, manage data sovereignty, coordinate pilot projects.	500,000
CP Logic	Project Management and Strategic Planning	Oversee project deliverables, Community Engagement support, Coordinate with all partners	300,000
Parachute Software	Platform Development and Integration	Build and customize the PAPER platform, integrate third-party tools, generate documentation templates.	900,000
Volta Research	Data Integration and Analysis	Customize Volta Snap, aggregate energy audit data, and create analysis modules.	400,000
University of Waterloo	Academic Validation and Capacity Building	Conduct validation studies, produce case studies, and provide scalability recommendations.	400,000
Legal Services	Contract Drafting, Licensing, and Data Sovereignty Policies	Draft contracts, develop the data sovereignty policy, and review agreements.	100,000
Miscellaneous Costs	Travel, contingency, and unforeseen expenses	Cover unexpected project needs. To be expanded on.	100,000

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9. Next Steps

1. **Refine Project Scope**:
    - Complete integrations of Volta Snap and CP Logic tools.

2. **Partner Cost Estimated**:
    - Align budget details, partner contributions, and pilot plans with grant criteria.
3. **Create and circulate draft grant application**:
    - Submit the finalized application to Prairies Economic Development Canada.

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10. Conclusion

PAPER represents a transformative opportunity for Indigenous communities to streamline energy retrofit planning while building internal capacity and fostering long-term sustainability. By automating critical processes and equipping local teams with AI-driven tools, PAPER addresses systemic barriers, reduces costs, and empowers communities to lead their own energy futures.